ESAT PRACTICE TEST SERIES

CHEMISTRY + BIOLOGY

LEVEL 0 · EASY

Chemistry + Biology

Mathematics 1 + Chemistry + Biology

Foundation-focused ESAT practice with more direct setup and shorter solution chains.

MODULE

Mathematics 1

MODULE

Chemistry

MODULE

Biology

FULL SOLUTION BOOK

ThrivingScholars

81 preserved questions · three verified module keys

COMBINED ANSWER KEYS

Chemistry + Biology · Level 0 Easy

This book compiles three existing 27-question modules into one 81-question pathway. No new questions have been generated and no source solution has been rewritten.

> **Composition:** Mathematics 1 + Chemistry + Biology. Each module retains its original question order, answer and worked solution.

Mathematics 1

Engineering Mock Test 1 · preserved Mathematics 1 module

Q1 G	Q2 C	Q3 B	Q4 B	Q5 E
Q6 G	Q7 D	Q8 C	Q9 B	Q10 C
Q11 A	Q12 D	Q13 B	Q14 A	Q15 D
Q16 C	Q17 B	Q18 D	Q19 C	Q20 E
Q21 E	Q22 C	Q23 E	Q24 D	Q25 E
Q26 C	Q27 C			

Chemistry

Preserved Chemistry Easy source set

Q1 B	Q2 D	Q3 C	Q4 D	Q5 D
Q6 D	Q7 F	Q8 C	Q9 B	Q10 H
Q11 C	Q12 A	Q13 D	Q14 C	Q15 B
Q16 A	Q17 F	Q18 B	Q19 C	Q20 B
Q21 C	Q22 D	Q23 B	Q24 C	Q25 C
Q26 E	Q27 C			

Biology

Preserved Biology Easy source set

Q1 C	Q2 A	Q3 H	Q4 C	Q5 E
Q6 E	Q7 C	Q8 D	Q9 B	Q10 B
Q11 E	Q12 G	Q13 C	Q14 D	Q15 B
Q16 B	Q17 E	Q18 E	Q19 E	Q20 E
Q21 D	Q22 A	Q23 B	Q24 C	Q25 B
Q26 H	Q27 B			

LEVEL 0 · EASY

Mathematics 1

27 QUESTIONS · COMPLETE SOLUTIONS

Engineering Mock Test 1 · preserved Mathematics 1 module. Question wording, option order, answer and solution method are unchanged.

M1-01

Mock Test 1 - Mathematics 1

Answer G

QUESTION 1

If the 5-digit number $1X2X3$ is divisible by 9, what is X?

- **A** 0
- **B** 1
- **C** 2
- **D** 3
- **E** 4
- **F** 5
- **G** 6

WORKED SOLUTION

The digit sum is $1 + X + 2 + X + 3 = 6 + 2X$. For divisibility by 9, $6 + 2X$ must be a multiple of 9. From the choices, $X = 6$ gives 18.

M1-02 Mock Test 1 - Mathematics 1 **Answer C**

QUESTION 2

If we define $a \star b = a^2 - b^2$, what is $5 \star 4$?

- **A** 1
- **B** 7
- **C** 9
- **D** 16
- **E** 25
- **F** 41

WORKED SOLUTION

Apply the definition directly: $5 \star 4 = 5^2 - 4^2 = 25 - 16 = 9$.

M1-03 Mock Test 1 - Mathematics 1 **Answer B**

QUESTION 3

Jack tosses two fair dice. What is the probability that the sum of the two values is no less than 9?

Recall that

$$\text{Probability} = \frac{\text{Number of valid cases}}{\text{Number of all possible cases}}.$$

A $\frac{1}{4}$

B $\frac{5}{18}$

C $\frac{1}{3}$

D $\frac{5}{12}$

E $\frac{2}{3}$

WORKED SOLUTION

For two dice, sums at least 9 are $9, 10, 11, 12$, giving $4+3+2+1=10$ outcomes. So the probability is $10/36=5/18$.

M1-04

Mock Test 1 - Mathematics 1

Answer B

QUESTION 4

How many 4-digit integers $\overline{abcd}$ satisfy that

1 The thousands digit a and the tens digit c are even numbers;

2 The hundreds digit b and the unit digit d are odd numbers.

3 This number is divisible by 5?

A 80

B 100

C 125

D 250

E 500

WORKED SOLUTION

Since the number is divisible by 5, the final digit must be 5 because d is odd. Then $a \in \{2, 4, 6, 8\}$, b has 5 choices, and c has 5 choices, giving $4 \cdot 5 \cdot 5 \cdot 1 = 100$.

M1-05

Mock Test 1 - Mathematics 1 | Answer E

QUESTION 5

If $\frac{m}{10} = \frac{6}{n} = \frac{2}{5}$, what is $m + n$?

A 10

B 13

C 14

D 16

E 19

WORKED SOLUTION

From $m/10 = 2/5$, $m = 4$. From $6/n = 2/5$, $30 = 2n$, so $n = 15$. Hence $m + n = 19$.

M1-06

Mock Test 1 - Mathematics 1

Answer G

QUESTION 6

Nancy has 100 marbles. She gives 20% of them to her friend Pedro. Then Nancy gives 25% of what is left to another friend, Ben. Finally, Nancy gives 10% of what is now left in the bag to her brother Jimmy. What percentage of her original bag of marbles does Nancy have left for herself?

- **A** 20
- **B** 30
- **C** $33\frac{1}{3}$
- **D** 38
- **E** 45
- **F** 50
- **G** 54
- **H** 60

WORKED SOLUTION

Track what remains: after 20% is given away, 80 remain; after giving 25% of 80, 60 remain; after giving 10% of 60, 54 remain.

M1-07

Mock Test 1 - Mathematics 1 | Answer D

QUESTION 7

What is the sum of all positive 2-digit integers that have a remainder of 1 when divided by 9 and a remainder of 2 when divided by 5?

A 29

B 56

C 84

D 119

E 164

WORKED SOLUTION

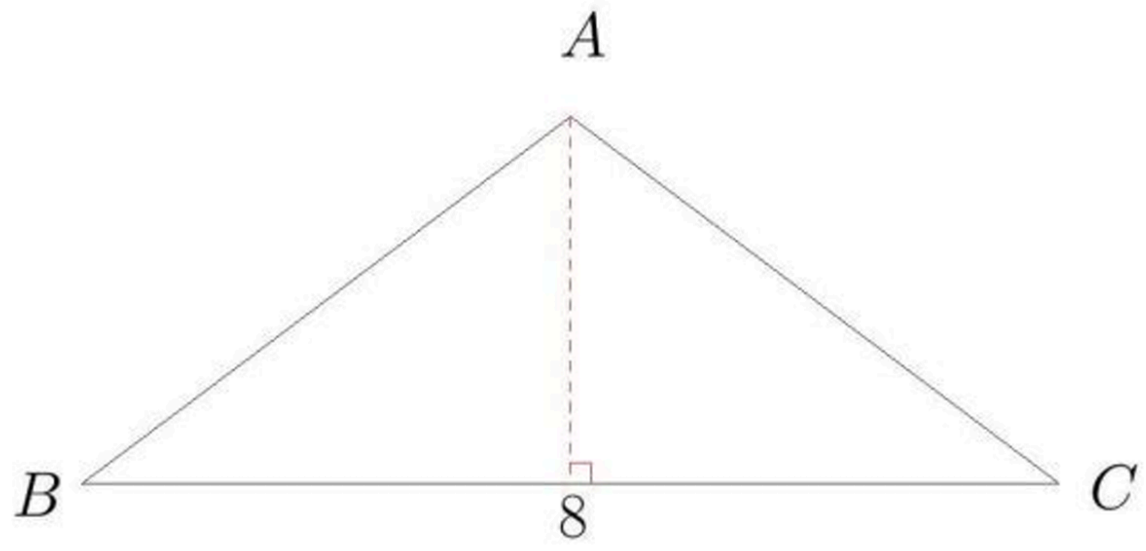
The numbers satisfying both conditions are 37 and 82: each is 1 more than a multiple of 9 and 2 more than a multiple of 5. Their sum is 119.

M1-08

Mock Test 1 - Mathematics 1 | Answer C

QUESTION 8

$\triangle ABC$ is an isosceles triangle. $AB = AC$. If $BC = 8$ and the area of $\triangle ABC$ is 12, what is the perimeter of $\triangle ABC$?



A 12

B 16

C 18

D 19

E 20

WORKED SOLUTION

The height is h where $\frac{1}{2} \cdot 8 \cdot h = 12$, so $h = 3$. The altitude bisects the base, giving half-base 4, so each equal side is $\sqrt{3^2 + 4^2} = 5$. Perimeter $= 8 + 5 + 5 = 18$.

M1-09

Mock Test 1 - Mathematics 1 | Answer B

QUESTION 9

What is the smaller angle between the hour hand and minute hand of a clock at $5{:}50$ pm?

- A 100°
- B 125°
- C 150°
- D 155°
- E 165°

WORKED SOLUTION

At $5{:}50$, the minute hand is at 300°. The hour hand is at $5 \cdot 30 + 50 \cdot 0.5 = 175^\circ$. The smaller angle is $300 - 175 = 125^\circ$.

M1-10

Mock Test 1 - Mathematics 1 | Answer C

QUESTION 10

The average age of 5 people in a room is 40 years. A 36-year-old person leaves the room. What is the average age of the four remaining people?

- A 39
- B 40

C 41

D 43

E 45

WORKED SOLUTION

The original total age is $5 \cdot 40 = 200$. After the 36-year-old leaves, the total is 164, so the new average is $164/4 = 41$.

M1-11

Mock Test 1 - Mathematics 1 | Answer A

QUESTION 11

In how many ways can we write 127 as the sum of two prime numbers?

A 0

B 1

C 2

D 3

E 4

WORKED SOLUTION

An odd number written as a sum of two primes must use 2 plus an odd prime. Here $127 = 2 + 125$, but 125 is not prime, so there are no ways.

M1-12

Mock Test 1 - Mathematics 1

Answer D

QUESTION 12

If for some positive integer n,

$$\left(1+\frac{1}{2}\right)\left(1+\frac{1}{3}\right)\left(1+\frac{1}{4}\right)\cdots\left(1+\frac{1}{n}\right)=50,$$

what is n?

A 20

B 39

C 65

D 99

E 100

WORKED SOLUTION

Each factor is $1+1/k=(k+1)/k$. The product telescopes: $\frac{3}{2}\cdot\frac{4}{3}\cdots\frac{n+1}{n}=\frac{n+1}{2}=50$. Thus $n=99$.

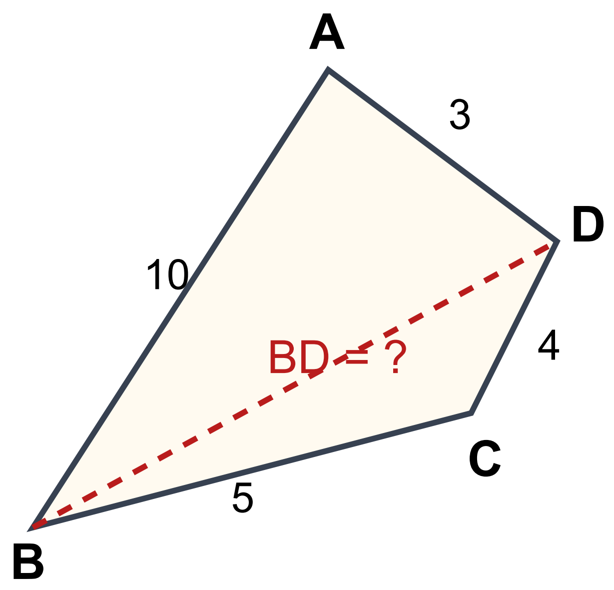
M1-13

Mock Test 1 - Mathematics 1

Answer B

QUESTION 13

In quadrilateral $ABCD$, $AB=10$, $BC=5$, $CD=4$ and $AD=3$. If the length of BD is an integer, what is BD?



A 7

B 8

C 9

D 10

E there is not enough information

WORKED SOLUTION

Use the triangle inequality in the two triangles that share BD. For $\triangle \mathrm{ABD}$, $|10-3| < \mathrm{BD} < 10+3$, so $7 < \mathrm{BD} < 13$. For $\triangle \mathrm{BCD}$, $|5-4| < \mathrm{BD} < 5+4$, so $1 < \mathrm{BD} < 9$. The only integer satisfying both is $\mathrm{BD} = 8$.

M1-14

Mock Test 1 - Mathematics 1 | Answer A

QUESTION 14

What is the unit digit of 7^{2020}? Here

$$7^n = \underbrace{7 \times 7 \times 7 \times \cdots \times 7}_{n}.$$

(Hint: check unit digits of $7, 7^2, 7^3, \ldots$ and find the pattern.)

- **A** 1
- **B** 3
- **C** 5
- **D** 7
- **E** 9

WORKED SOLUTION

The units digits of powers of 7 cycle as $7, 9, 3, 1$. Since 2020 is divisible by 4, the units digit is the fourth term of the cycle, 1.

M1-15

Mock Test 1 - Mathematics 1 | Answer D

QUESTION 15

Here x, y, z and u represent four different numbers.

$$\begin{array}{rr} & x\;y \\ + & z\;x \\ \hline & u\;x \end{array} \qquad \begin{array}{rr} & x\;y \\ - & z\;x \\ \hline & x \end{array}$$

$x + y + z + u =$

A 12

B 14

C 16

D 18

E 20

WORKED SOLUTION

From the subtraction, $(10x + y) - (10z + x) = x$, so $8x + y = 10z$. From the addition, $(10x + y) + (10z + x) = 10u + x$, so $10x + y + 10z = 10u$. The distinct digit solution is $(x, y, z, u) = (5, 0, 4, 9)$, giving 18.

M1-16

Mock Test 1 - Mathematics 1

Answer C

QUESTION 16

Billy, a seasonal worker in the town of Cowra, collected 8 buckets of cherries on his first day. Each day after that he increased the number of buckets he picked by 2 buckets per day. According to this pattern, how many buckets will he collect in the first 20 days?

A 308

B 500

C 540

D 580

E 600

WORKED SOLUTION

The buckets form an arithmetic sequence: $8, 10, 12, \ldots$. The 20th term is $8 + 19 \cdot 2 = 46$. Sum $= \frac{20}{2}(8 + 46) = 540$.

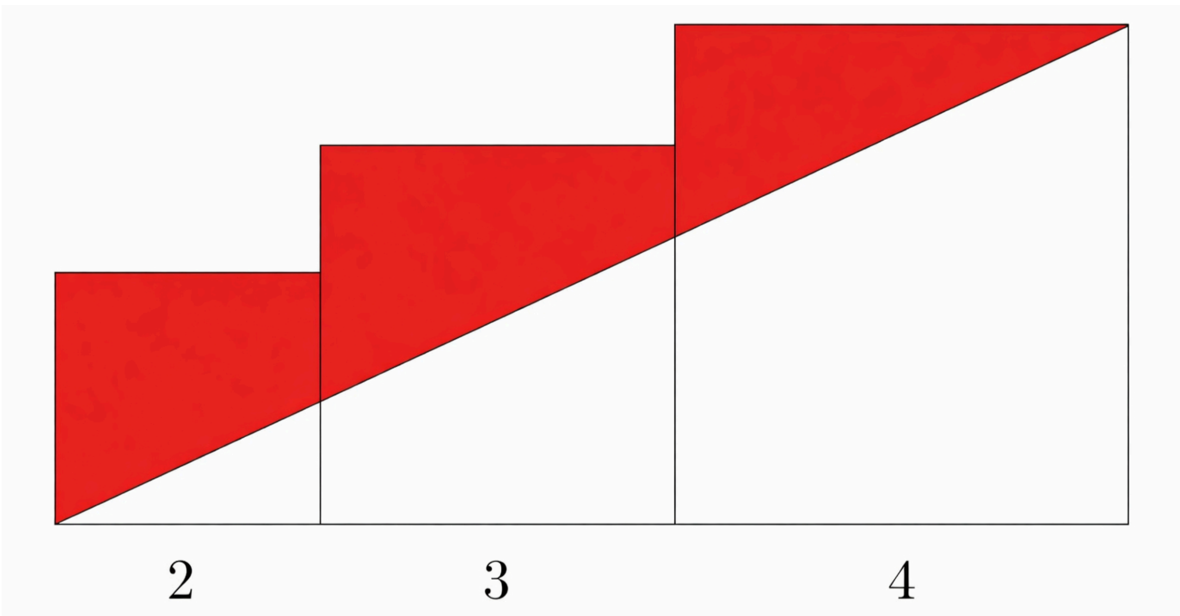
M1-17

Mock Test 1 - Mathematics 1

Answer B

QUESTION 17

Three squares are lined up horizontally as shown. Their side lengths are 2, 3 and 4 respectively. A straight line is drawn from the top right corner of the largest square to the bottom left hand corner of the smallest square. What is the area of the shaded region?



A 10

B 11

C 13

D 16

E 18

WORKED SOLUTION

The total area of the three squares is $2^2 + 3^2 + 4^2 = 29$. The line cuts off one large triangle of base 9 and height 4, area 18. Shaded area $= 29 - 18 = 11$.

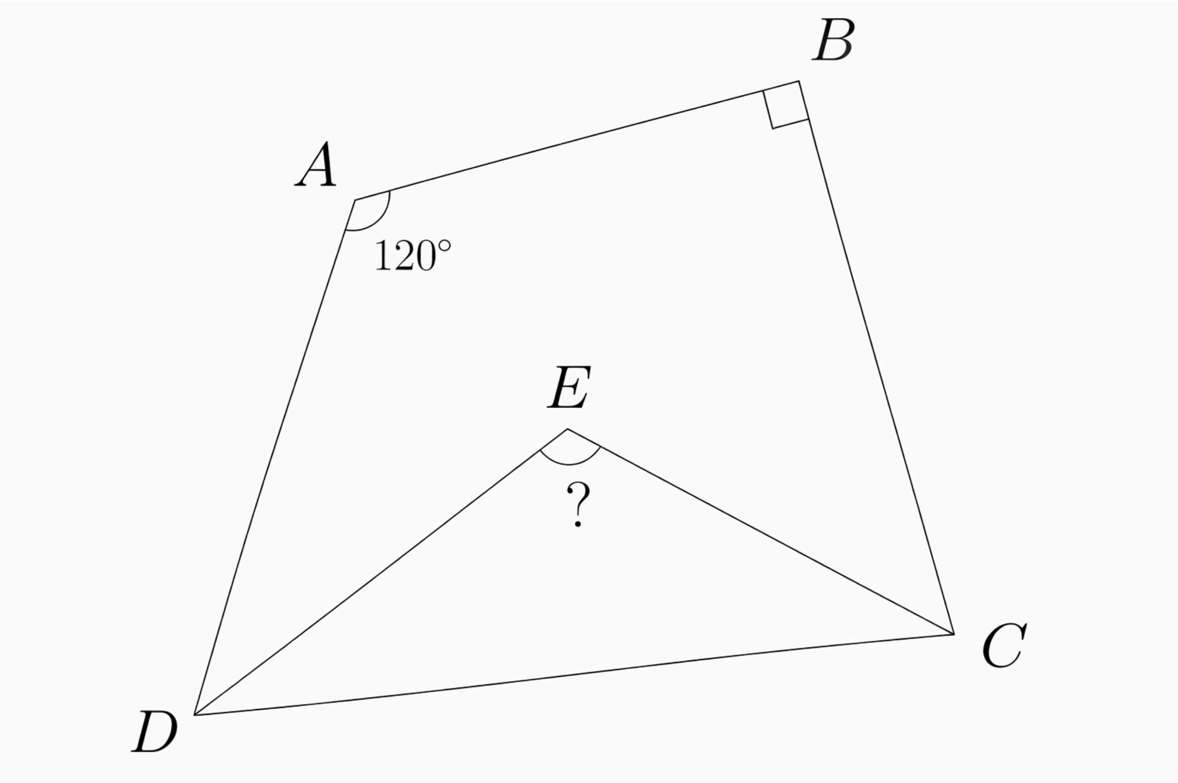
M1-18

Mock Test 1 - Mathematics 1

Answer **D**

QUESTION 18

In quadrilateral $ABCD$, $\angle A = 120°$ and $\angle B = 90°$. ED and EC bisect $\angle ADC$ and $\angle BCD$ respectively. What is $\angle E$?



A 90

B 95

C 100

D 105

E 110

WORKED SOLUTION

In the quadrilateral, $\angle C + \angle D = 360^\circ - 120^\circ - 90^\circ = 150^\circ$. Since EC and ED bisect those angles, triangle DCE has angles at C, D summing to 75°. Therefore $\angle E = 180^\circ - 75^\circ = 105^\circ$.

M1-19

Mock Test 1 - Mathematics 1

Answer C

QUESTION 19

In a mathematics contest with 10 problems, a student gains 6 points for a correct answer and loses 1 point for an incorrect answer. If Olivia answered every problem and her score was 32, how many correct answers did she have?

A 4

B 5

C 6

D 7

E 8

WORKED SOLUTION

Let c be the number correct. Then incorrect answers are $10 - c$, so $6c - (10 - c) = 32$. Hence $7c = 42$ and $c = 6$.

M1-20

Mock Test 1 - Mathematics 1

Answer E

QUESTION 20

Emily and Owen are going to a library that is 1 mile from their house. They leave home simultaneously. Emily rides her bicycle to the library at a constant speed of 12 miles per hour. Owen walks to the library at a constant speed of 5 miles per hour. How many minutes before Owen does Emily arrive?

A 3

B 4

C 5

D 6

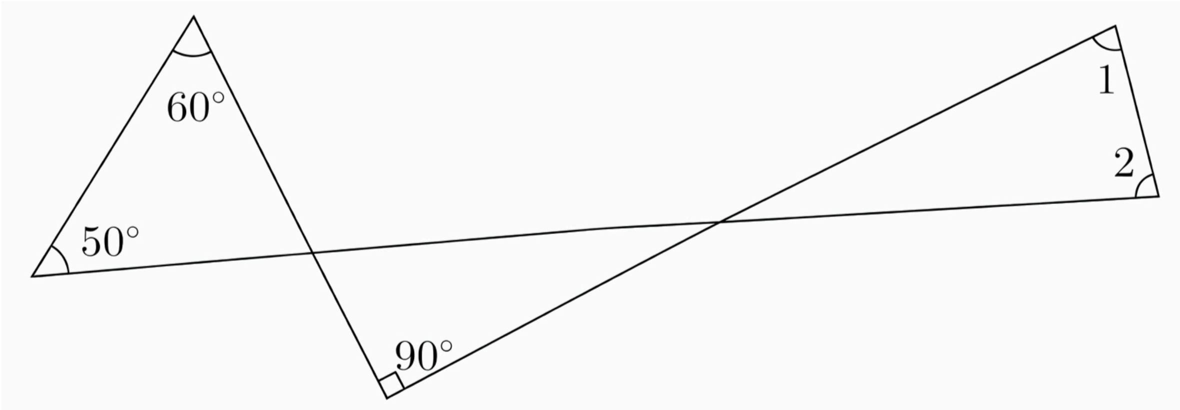
E 7

WORKED SOLUTION

Emily takes $1/12$ hour $= 5$ minutes. Owen takes $1/5$ hour $= 12$ minutes. Emily arrives $12 - 5 = 7$ minutes earlier.

QUESTION 21

In the following picture, if $\angle 1 = \angle 2$, what is $\angle 1$?



A 60°

B 65°

C 70°

D 75°

E 80°

WORKED SOLUTION

The left triangle gives the angle at the first intersection as $180^\circ - 50^\circ - 60^\circ = 70^\circ$. The next triangle has a right angle, so the angle at the second intersection is 20°. The final small triangle has angles $x, x, 20^\circ$, so $2x = 160^\circ$, giving $x = 80^\circ$.

M1-22

Mock Test 1 - Mathematics 1 | Answer C

QUESTION 22

Jack, a zoo keeper, is going to give N peaches to a group of monkeys. If he gives 10 peaches to each monkey, two monkeys will receive no peach. Or, the peaches are exactly enough to give 8 to each monkey. What is the sum of digits of N?

A 5

B 6

C 8

D 9

E 10

WORKED SOLUTION

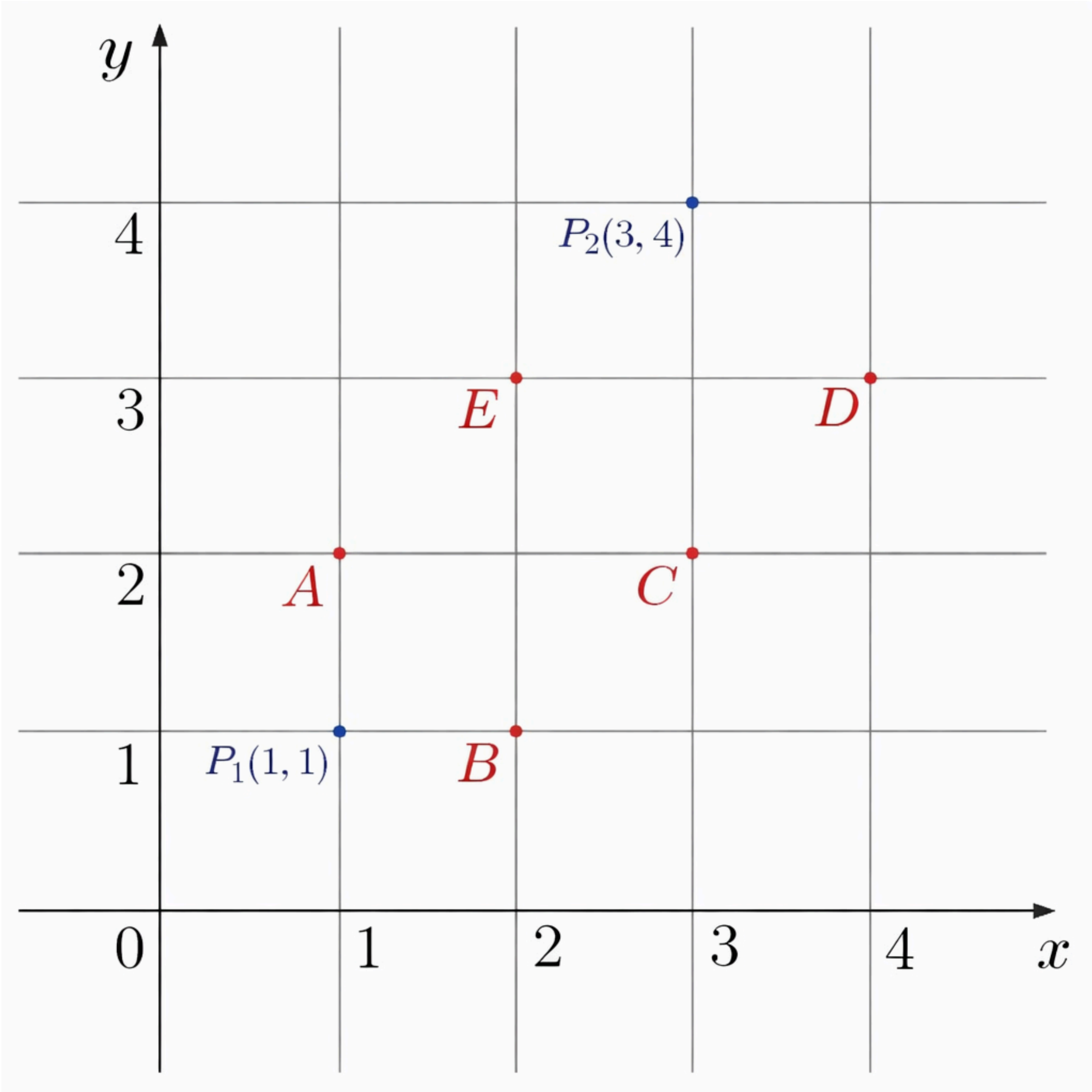
Let there be m monkeys. The two descriptions give $N = 8m$ and $N = 10(m - 2)$. Thus $8m = 10m - 20$, so $m = 10$ and $N = 80$. Digit sum $= 8$.

M1-23

Mock Test 1 - Mathematics 1 | Answer E

QUESTION 23

Every point on the plane can be assigned with a coordinate that has two numbers. For example, the coordinate of point P_1 in the following picture is $(1, 1)$ and the coordinate of point P_2 is $(3, 4)$. Which of the following points has the coordinate $(2, 3)$?



A A

B B

C C

D D

E E

WORKED SOLUTION

Read the grid directly: the point at $(2, 3)$ is labelled E.

M1-24

Mock Test 1 - Mathematics 1

Answer **D**

QUESTION 24

A pizza-shop offers a basic version of pizza with mozzarella and tomatoes. Two different toppings from the following list must be added: anchovies, artichokes, mushrooms, capers. Moreover, for each pizza three different sizes are available: small, medium, large. How many different types of pizza are available at all?

A 11

B 12

C 15

D 18

E 36

WORKED SOLUTION

Choose 2 toppings from 4: $\binom{4}{2} = 6$. For each, there are 3 sizes, so total pizzas $= 6 \cdot 3 = 18$.

M1-25

Mock Test 1 - Mathematics 1

Answer **E**

QUESTION 25

Bag A: 1 Snickers bar + 1 Butterfinger bar.

Bag B: 1 Butterfinger bar + 3 Reese's Peanut Butter Cups.

Bag C: 1 Snickers bar + 2 Reese's Peanut Butter Cups.

If each bag costs 50 cents, what is the price of
1 Snickers bar + 1 Butterfinger bar + 1 Reese's Peanut Butter Cup,
in cents?

A 40

B 45

C 50

D 55

E 60

WORKED SOLUTION

Let the prices be S, B, R. The bags give $S + B = 50$, $B + 3R = 50$, and $S + 2R = 50$. Solving gives $S = 30$, $B = 20$, $R = 10$, so $S + B + R = 60$.

M1-26 Mock Test 1 - Mathematics 1 Answer **C**

QUESTION 26

For how many positive integers x is

$$(x - 1)(x - 4)(x - 9)$$

negative?

A 2

B 3

C 4

D 5

E 6

WORKED SOLUTION

The roots are 1, 4 and 9. Testing the sign in the intervals shows the product is negative for $x < 1$ and $4 < x < 9$. The positive integers satisfying this are $5, 6, 7, 8$, so there are 4 values.

M1-27

Mock Test 1 - Mathematics 1 | Answer C

QUESTION 27

A gumball machine contains 9 red, 7 white, and 8 blue gumballs. The least number of gumballs a person must buy to be sure of getting four gumballs of the same color is

A 8

B 9

C 10

D 12

E 18

WORKED SOLUTION

The most gumballs possible without having four of one colour is $3 + 3 + 3 = 9$. One more guarantees four of some colour, so the least number is 10.

LEVEL 0 · EASY

Chemistry

27 QUESTIONS · COMPLETE SOLUTIONS

Preserved Chemistry Easy source set. Question wording, option order, answer and solution method are unchanged.

C-01

Periodic Table and electron configuration

Answer B

QUESTION 1

Four atoms have the following electron configurations.

atom	electron configuration
P	2,8,8
Q	2,8,1
R	2,8,7
S	2,8,8,1

Which statement is correct?

A P is a Period 2 noble gas.

B Q and S are in Group 1, and S is expected to react more vigorously with water than Q.

C R normally forms R^{2-} ions.

D P has one electron in its outer shell.

E Q is less metallic than R.

WORKED SOLUTION

Q has electron configuration 2,8,1 and S has electron configuration 2,8,8,1, so both are in Group 1.

Group 1 reactivity increases down the group. S has one more occupied electron shell than Q, so S is below Q and is expected to react more vigorously with water.

P has three occupied shells and a full outer shell, so it is a Period 3 noble gas. R has seven outer-shell electrons and normally forms a 1− ion.

Therefore the correct answer is **B**.

C-02

Formulae of ionic compounds

Answer **D**

QUESTION 2

Aluminium ions are Al^{3+} and sulfate ions are SO_4^{2-}.

Which row gives the formula of aluminium sulfate and the number of sulfate ions in one formula unit?

option	formula	sulfate ions
A	$AlSO_4$	1
B	$Al(SO_4)_2$	2
C	Al_2SO_4	1
D	$Al_2(SO_4)_3$	3
E	$Al_3(SO_4)_2$	2

A A

B B

C C

D D

E E

WORKED SOLUTION

The smallest common magnitude of charge is 6:

$$2 \times (+3) = +6, \qquad 3 \times (-2) = -6.$$

Therefore two aluminium ions combine with three sulfate ions:

$$\boxed{Al_2(SO_4)_3}.$$

There are three sulfate ions in one formula unit, so the correct row is **D**.

C-03 Ionic equations Answer **C**

QUESTION 3

Aqueous magnesium chloride is mixed with aqueous sodium carbonate. A white precipitate of magnesium carbonate forms.

Which is the ionic equation for this reaction?

A $Na^+(aq) + Cl^-(aq) \rightarrow NaCl(s)$

B $Mg(aq) + CO_3(aq) \rightarrow MgCO_3(s)$

C $Mg^{2+}(aq) + CO_3^{2-}(aq) \rightarrow MgCO_3(s)$

D $2Na^+(aq) + CO_3^{2-}(aq) \rightarrow Na_2CO_3(s)$

E $Mg^{2+}(aq) + 2Cl^-(aq) \rightarrow MgCl_2(s)$

WORKED SOLUTION

The sodium and chloride ions remain dissolved and are spectator ions.

The ions that actually form the precipitate are magnesium ions and carbonate ions:

$$Mg^{2+}(aq) + CO_3^{2-}(aq) \rightarrow MgCO_3(s).$$

Therefore the answer is **C**.

C-04

Moles and particles

Answer **D**

QUESTION 4

How many moles of helium atoms contain the same total number of **atoms** as 0.20 mol of oxygen molecules, O_2?

A $0.10\ \text{mol}$

B $0.20\ \text{mol}$

C $0.25\ \text{mol}$

D $0.40\ \text{mol}$

E $0.80\ \text{mol}$

WORKED SOLUTION

Each oxygen molecule contains two oxygen atoms. Therefore 0.20 mol of O_2 contains

$$2(0.20) = 0.40\ \text{mol}$$

of atoms.

Helium is monatomic, so 0.40 mol of helium contains the same number of atoms.

The answer is **D**.

C-05

Purity

Answer D

QUESTION 5

A $2.50\ \mathrm{g}$ impure sample of calcium carbonate contains impurities that do not react with hydrochloric acid.

The sample is reacted with excess hydrochloric acid and produces $0.480\ \mathrm{dm}^3$ of carbon dioxide at room temperature and pressure.

At rtp,

$$1\ \text{mol of gas} = 24.0\ \mathrm{dm}^3.$$

$$A_r: \quad Ca = 40,\ C = 12,\ O = 16$$

What is the percentage purity of the calcium carbonate?

A 40%

B 60%

C 75%

D 80%

E 90%

WORKED SOLUTION

The amount of carbon dioxide formed is

$$n(CO_2) = \frac{0.480}{24.0} = 0.0200\ \text{mol}.$$

From

$$CaCO_3 + 2HCl \rightarrow CaCl_2 + CO_2 + H_2O,$$

the sample contained 0.0200 mol of calcium carbonate.

$$M_r(CaCO_3) = 40 + 12 + 3(16) = 100.$$

So the mass of pure calcium carbonate was

$$0.0200 \times 100 = 2.00 \text{ g}.$$

Hence

$$\text{purity} = \frac{2.00}{2.50} \times 100 = 80\%.$$

The answer is **D**.

C-06

Kinetic particle theory

Answer **D**

QUESTION 6

A pure substance is heated. During melting, energy continues to be supplied but its temperature remains constant until all of the solid has melted.

Which statement best explains this?

A The particles stop moving while the substance melts.

B The supplied energy breaks the atoms into smaller particles.

C The supplied energy increases the average kinetic energy of the particles, but the temperature does not change.

D The supplied energy is used in overcoming forces between particles rather than increasing their average kinetic energy.

E Particles in the solid and liquid have exactly the same arrangement.

WORKED SOLUTION

Temperature is related to the average kinetic energy of the particles.

During melting the temperature is constant, so the average kinetic energy is not increasing. The supplied energy is instead used to overcome forces holding particles in the solid structure.

Therefore the correct answer is **D**.

C-07

Metallic bonding

Answer F

QUESTION 7

Consider the following statements about a solid metal.

1. Some electrons are delocalised and can move through the structure.
2. Electrical conduction occurs because positive metal ions move through the solid.
3. Layers of metal ions can move relative to one another while metallic bonding is maintained.

Which statements are correct?

- **A** none
- **B** 1 only
- **C** 2 only
- **D** 3 only
- **E** 1 and 2 only
- **F** 1 and 3 only
- **G** 2 and 3 only
- **H** 1, 2 and 3

WORKED SOLUTION

Statement 1 is correct: metals contain delocalised electrons that can move through the structure.

Statement 2 is false: the positive metal ions are fixed in the solid lattice; electrical conduction is due to mobile delocalised electrons.

Statement 3 is correct: layers can slide while attraction between positive ions and delocalised electrons is maintained.

So statements 1 and 3 only are correct. The answer is **F**.

C-08

Giant structures

Answer **C**

QUESTION 8

Which substance has a very high melting point, does not conduct electricity when solid, and does not become an electrical conductor simply on melting?

- **A** sodium chloride
- **B** graphite
- **C** silicon dioxide
- **D** copper
- **E** methane

WORKED SOLUTION

Silicon dioxide has a giant covalent structure. Many strong covalent bonds must be overcome to melt it, so its melting point is very high.

It does not have mobile ions or delocalised electrons available for electrical conduction, whether solid or molten.

Sodium chloride is different because its ions become mobile when molten.

Therefore the answer is **C**.

C-09

Group 1 trends

Answer **B**

QUESTION 9

Which row correctly shows the trends from lithium to sodium to potassium?

option	reactivity with water	melting point
A	$Li > Na > K$	$Li > Na > K$
B	$K > Na > Li$	$Li > Na > K$
C	$K > Na > Li$	$K > Na > Li$
D	$Li > Na > K$	$K > Na > Li$
E	$Na > K > Li$	$Li > K > Na$

A A

B B

C C

D D

E E

WORKED SOLUTION

Group 1 metals become more reactive down the group:

$$K > Na > Li.$$

Their melting points decrease down the group:

$$Li > Na > K.$$

Only row **B** gives both trends correctly.

C-10

Air and drinking water

Answer **H**

QUESTION 10

Consider these statements.

1. Dry air contains approximately 78% nitrogen.
2. Chlorine may be added during drinking-water treatment to kill harmful microorganisms.
3. Fluoride ions may be added to drinking water to help reduce tooth decay.
4. Components of air can be separated by fractional distillation after the air has been liquefied.

Which statements are correct?

A 1 and 2 only

B 1 and 3 only

C 2 and 4 only

D 1, 2 and 3 only

E 1, 2 and 4 only

F 1, 3 and 4 only

G 2, 3 and 4 only

H 1, 2, 3 and 4

WORKED SOLUTION

All four statements are correct.

Dry air is about 78% nitrogen. Chlorine is used for disinfection in water treatment, fluoride ions may be added to help reduce tooth decay, and the major gases in air can be separated by fractional distillation after liquefaction.

Therefore the answer is **H**.

C-11

Separation techniques

Answer **C**

QUESTION 11

A mixture contains:

- insoluble sand,
- hexane,
- water containing dissolved sodium chloride.

Hexane and water are immiscible.

Which sequence is suitable for first removing the sand, then isolating the hexane, and finally obtaining sodium chloride crystals?

A separating funnel → filtration → evaporation

B filtration → crystallisation → separating funnel

C filtration → separating funnel → crystallisation

D crystallisation → filtration → separating funnel

E centrifugation → chromatography → filtration

WORKED SOLUTION

First remove the insoluble sand by filtration.

The filtrate contains two immiscible liquid layers, hexane and aqueous sodium chloride, so a separating funnel can isolate the hexane.

Finally, sodium chloride crystals can be obtained from the aqueous layer by crystallisation.

The correct sequence is therefore **filtration** → **separating funnel** → **crystallisation**, answer **C**.

C-12

Obtaining a pure solvent

Answer **A**

QUESTION 12

A mixture contains water, dissolved sodium chloride, and a small amount of insoluble charcoal.

Which procedure is most suitable for obtaining a sample of **pure water** from the mixture?

- **A** Filter, then use simple distillation on the filtrate.
- **B** Filter, then evaporate the filtrate to dryness.
- **C** Use a separating funnel, then crystallise.
- **D** Use fractional distillation without first filtering.
- **E** Use paper chromatography.

WORKED SOLUTION

The insoluble charcoal should first be removed by filtration.

The filtrate is salt water. Sodium chloride is non-volatile, while water can be vaporised and condensed, so simple distillation gives pure water as the distillate.

Therefore the answer is **A**.

C-13

pH scale

Answer **D**

QUESTION 13

A strong acid has pH 2.

It is diluted with water until the total volume is 100 times its original volume. Assume the acid remains fully ionised.

What is its new pH?

A 1

B 2

C 3

D 4

E 6

WORKED SOLUTION

A 100-fold dilution reduces the hydrogen-ion concentration by a factor of

$$100 = 10^2.$$

Each factor of 10 reduction in $[H^+]$ increases pH by 1, so the pH increases by 2:

$$2 + 2 = 4.$$

Therefore the answer is **D**.

C-14

Polyprotic acids

Answer **C**

QUESTION 14

$20.0\ \mathrm{cm}^3$ of $0.30\ \mathrm{mol\ dm}^{-3}$ sulfuric acid is completely neutralised by $0.50\ \mathrm{mol\ dm}^{-3}$ sodium hydroxide.

Assume one mole of sulfuric acid supplies two moles of H^+.

What volume of sodium hydroxide is required?

- **A** $12.0\ \mathrm{cm}^3$
- **B** $18.0\ \mathrm{cm}^3$
- **C** $24.0\ \mathrm{cm}^3$
- **D** $30.0\ \mathrm{cm}^3$
- **E** $48.0\ \mathrm{cm}^3$

WORKED SOLUTION

The amount of sulfuric acid is

$$n(H_2SO_4) = 0.30 \times 0.0200 = 0.00600\ \mathrm{mol}.$$

Each mole supplies two moles of H^+, so 0.0120 mol of OH^- is needed.

Therefore

$$V(NaOH) = \frac{0.0120}{0.50} = 0.0240\ \mathrm{dm}^3 = 24.0\ \mathrm{cm}^3.$$

The answer is **C**.

C-15

Preparation of salts

Answer **B**

QUESTION 15

A student wants to prepare pure, dry crystals of copper(II) sulfate using dilute sulfuric acid and insoluble copper(II) oxide.

Which method is best?

- **A** Add exactly one spatula of copper oxide to cold acid and evaporate immediately to dryness.
- **B** Warm the acid, add copper oxide until some remains unreacted, filter, partially evaporate the filtrate, cool it, then filter and dry the crystals.
- **C** Add an indicator to the acid and titrate it with solid copper oxide.
- **D** Dissolve the copper oxide in water, filter, then add sulfuric acid.
- **E** Mix the reagents and collect the product by simple distillation.

WORKED SOLUTION

Copper(II) oxide is insoluble. Adding it in excess ensures that all the sulfuric acid has reacted:

$$\mathrm{CuO} + \mathrm{H_2SO_4} \rightarrow \mathrm{CuSO_4} + \mathrm{H_2O}.$$

Unreacted copper oxide is removed by filtration. The filtrate is then partially evaporated and cooled to crystallise copper(II) sulfate, after which the crystals are filtered and dried.

Therefore the answer is **B**.

C-16

Measuring reaction rate

Answer **A**

QUESTION 16

Magnesium carbonate reacts with excess dilute hydrochloric acid.

$$MgCO_3 + 2HCl \rightarrow MgCl_2 + CO_2 + H_2O$$

Which measurement could be used to follow the rate of the reaction **without collecting the carbon dioxide gas**?

A Measure the total mass of the reaction vessel and contents at regular time intervals.

B Measure only the final pH.

C Measure the original volume of hydrochloric acid.

D Measure the mass of magnesium chloride before the reaction begins.

E Measure only the room temperature.

WORKED SOLUTION

Carbon dioxide leaves the reaction vessel as the reaction proceeds, so the total mass of the vessel and contents decreases.

Recording the mass at regular time intervals therefore provides a measure of the reaction progress and hence its rate.

The answer is **A**.

C-17

Factors affecting rate

Answer F

QUESTION 17

Three experiments use the same mass of calcium carbonate and excess hydrochloric acid at the same temperature.

experiment	calcium carbonate	acid concentration
1	chips	$1.0\ \mathrm{mol\ dm^{-3}}$
2	powder	$1.0\ \mathrm{mol\ dm^{-3}}$
3	chips	$2.0\ \mathrm{mol\ dm^{-3}}$

The same volume of acid is used each time.

Consider the statements.

1. Experiment 2 should reach completion sooner than experiment 1.
2. Experiment 3 should produce twice the final volume of carbon dioxide as experiment 1.
3. Experiments 2 and 3 should both have greater initial rates than experiment 1.

Which statements are correct?

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Powder has a larger surface area than chips, so experiment 2 has a greater rate and should reach completion sooner than experiment 1. Statement 1 is true.

Experiment 3 has a higher acid concentration, so it also has a greater initial rate. Statement 3 is true.

However, the same mass of calcium carbonate is used and the acid is in excess in every experiment. Calcium carbonate therefore determines the total amount of carbon dioxide, so the final gas volume is the same. Statement 2 is false.

Thus statements 1 and 3 only are correct: **F**.

C-18 Electrolysis and half-equations **Answer B**

QUESTION 18

Molten magnesium bromide is electrolysed using inert electrodes.

Which row correctly explains why **direct current** is used and gives the half-equation at the **anode**?

option	reason for using dc	anode half-equation
A	keeps electrode polarity fixed	$Mg^{2+} + 2e^{-} \rightarrow Mg$
B	keeps electrode polarity fixed	$2Br^{-} \rightarrow Br_2 + 2e^{-}$
C	continually reverses electrode polarity	$2Br^{-} \rightarrow Br_2 + 2e^{-}$
D	continually reverses electrode polarity	$Mg^{2+} + 2e^{-} \rightarrow Mg$
E	prevents ions moving	$Br_2 + 2e^{-} \rightarrow 2Br^{-}$

A A

B B

C C

D D

E E

WORKED SOLUTION

Direct current keeps each electrode at a fixed polarity. With alternating current the electrode polarity would repeatedly reverse.

At the positive electrode, bromide ions are oxidised:

$$2Br^- \rightarrow Br_2 + 2e^-.$$

Therefore row **B** is correct.

C-19

Hydrocarbon properties

Answer **C**

QUESTION 19

As the carbon-chain length of straight-chain alkanes increases from C_4 to C_8, which combination of trends is expected?

option	boiling point	viscosity	flammability
A	decreases	decreases	decreases
B	increases	decreases	increases
C	increases	increases	decreases
D	decreases	increases	decreases
E	increases	decreases	decreases

A A

B B

C C

D D

E E

WORKED SOLUTION

As chain length increases, intermolecular attractions become stronger.

Therefore boiling point and viscosity increase. Larger hydrocarbons are less volatile and generally less readily flammable.

The correct combination is **increase, increase, decrease**, so the answer is **C**.

C-20

Structural isomerism

Answer B

QUESTION 20

Which pair consists of structural isomers?

A CH_4 and C_2H_6

B $CH_3CH_2CH_2CH_3$ and $CH_3CH(CH_3)CH_3$

C $CH_2{=}CH_2$ and CH_3CH_3

D CH_3CH_2OH and CH_3COOH

E $CH_3CH_2CH_3$ and $CH_3CH_2CH_2CH_3$

WORKED SOLUTION

Structural isomers have the same molecular formula but different structural formulae.

Both butane, $CH_3CH_2CH_2CH_3$, and methylpropane, $CH_3CH(CH_3)CH_3$, have molecular formula C_4H_{10}, but the atoms are connected differently.

Therefore the answer is **B**.

C-21 Cracking Answer **C**

QUESTION 21

A hydrocarbon undergoes cracking:

$$C_{12}H_{26} \rightarrow C_8H_{18} + X.$$

What is X?

- **A** C_2H_4
- **B** C_3H_6
- **C** C_4H_8
- **D** C_4H_{10}
- **E** C_5H_{10}

WORKED SOLUTION

Balance the carbon atoms:

$$12 = 8 + 4,$$

so X contains four carbon atoms.

Balance the hydrogen atoms:

$$26 = 18 + 8.$$

Therefore $X = C_4H_8$, which is consistent with cracking producing an alkene.

The answer is **C**.

C-22

Reactions of alkenes

Answer **D**

QUESTION 22

Ethene is shaken with bromine water.

Which row correctly gives the observation and the organic product?

option	observation	organic product
A	bromine colour remains	C_2H_6
B	bromine water decolourises	C_2H_6
C	bromine water decolourises	C_2H_5Br
D	bromine water decolourises	$C_2H_4Br_2$
E	solution turns blue	$C_2H_4Br_2$

A A

B B

C C

D D

E E

WORKED SOLUTION

Bromine adds across the carbon-carbon double bond in ethene:

$$CH_2{=}CH_2 + Br_2 \rightarrow CH_2BrCH_2Br.$$

The bromine-water colour disappears and the product has molecular formula $C_2H_4Br_2$.

Therefore row **D** is correct.

C-23

Condensation polymers

Answer **B**

QUESTION 23

A diol,

$$HO-CH_2-CH_2-OH,$$

reacts with a dicarboxylic acid,

$$HOOC-CH_2-COOH.$$

Which row correctly identifies the type of polymer formed and the small molecule eliminated during polymerisation?

A polyester; hydrogen

B polyester; water

C polyamide; water

D addition polymer; no small molecule

E polyamide; carbon dioxide

WORKED SOLUTION

A diol reacting with a dicarboxylic acid forms ester links, $-COO-$, between monomer units.

The polymer is therefore a polyester. Each condensation step eliminates a molecule of water.

The answer is **B**.

C-24

Alcohols | Answer C

QUESTION 24

Ethanol reacts with excess sodium.

$$2C_2H_5OH + 2Na \rightarrow 2C_2H_5ONa + H_2$$

What volume of hydrogen gas at rtp is formed when 4.60 g of ethanol reacts completely?

$$M_r(C_2H_5OH) = 46$$

and

$$1 \text{ mol gas} = 24.0 \text{ dm}^3.$$

A 0.60 dm^3

B 0.80 dm^3

C 1.20 dm^3

D 2.40 dm^3

E 4.80 dm^3

WORKED SOLUTION

The amount of ethanol is

$$n = \frac{4.60}{46} = 0.100 \text{ mol}.$$

From the equation, two moles of ethanol produce one mole of hydrogen, so

$$n(H_2) = 0.0500 \text{ mol}.$$

At rtp,

$$V = 0.0500 \times 24.0 = 1.20 \text{ dm}^3.$$

The answer is **C**.

C-25

Carboxylic acids

Answer **C**

QUESTION 25

Ethanoic acid reacts completely with calcium carbonate.

Which pair gives the salt formed and the gas released?

- **A** $CaCO_3$; H_2
- **B** $Ca(CH_3COO)$; CO_2
- **C** $Ca(CH_3COO)_2$; CO_2
- **D** $Ca(CH_3COO)_2$; H_2
- **E** CH_3COOCa; O_2

WORKED SOLUTION

An acid reacting with a carbonate produces a salt, carbon dioxide and water.

The ethanoate ion is CH_3COO^- and calcium is Ca^{2+}, so the salt is $Ca(CH_3COO)_2$.

The balanced reaction is

$$2CH_3COOH + CaCO_3 \rightarrow Ca(CH_3COO)_2 + CO_2 + H_2O.$$

Therefore the answer is **C**.

C-26

Metals and extraction | Answer E

QUESTION 26

Consider these statements.

1. A metal that is more reactive than carbon cannot normally be extracted from its oxide by reduction with carbon.
2. Compounds of transition metals are often coloured.
3. Transition metals form ions in only one oxidation state.

Which statements are correct?

- **A** none
- **B** 1 only
- **C** 2 only
- **D** 3 only
- **E** 1 and 2 only
- **F** 1 and 3 only
- **G** 2 and 3 only
- **H** 1, 2 and 3

WORKED SOLUTION

Statement 1 is correct: metals above carbon in the reactivity series cannot normally be extracted from their oxides using carbon reduction.

Statement 2 is correct: coloured compounds are a common characteristic of transition metals.

Statement 3 is false: transition metals commonly form ions in more than one oxidation state.

Therefore statements 1 and 2 only are correct, so the answer is **E**.

C-27

Chemical tests

Answer C

QUESTION 27

An unknown aqueous salt gives:

- a **crimson-red flame**;
- a **white precipitate** when aqueous barium chloride is added in the presence of dilute hydrochloric acid.

Which salt is the most likely identity of the unknown?

A $LiCl$

B Na_2SO_4

C Li_2SO_4

D K_2SO_4

E $CuSO_4$

F $LiNO_3$

WORKED SOLUTION

A crimson-red flame identifies lithium ions, Li^+.

A white precipitate with acidified barium chloride indicates sulfate ions, SO_4^{2-}, because barium sulfate is insoluble and white.

The salt must therefore be Li_2SO_4.

The answer is **C**.

LEVEL 0 · EASY

Biology

27 QUESTIONS · COMPLETE SOLUTIONS

Preserved Biology Easy source set. Question wording, option order, answer and solution method are unchanged.

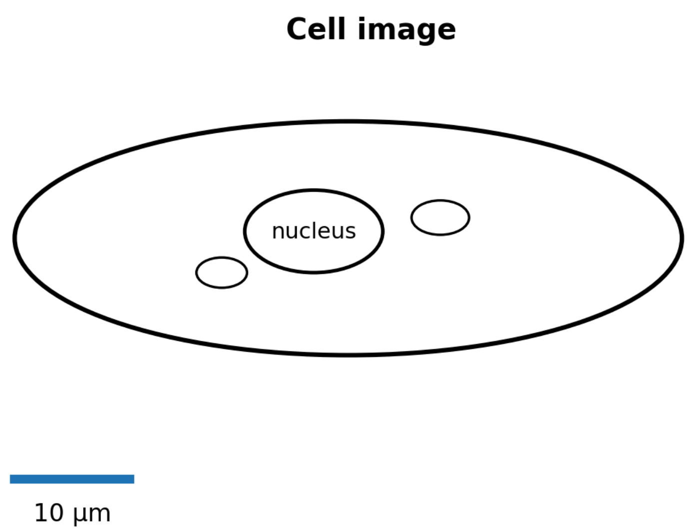
B-01

Microscopy

Answer C

QUESTION 1

The diagram shows a cell viewed using a microscope. The scale bar represents **10 µm**.



Approximately how long is the cell along its longest axis?

A 20 µm

B 40 µm

C 60 μm

D 80 μm

E 100 μm

WORKED SOLUTION

The cell is approximately 5.4 scale-bar lengths long.

$$5.4 \times 10\ \mu\text{m} \approx 54\ \mu\text{m}$$

The nearest option is **60 μm**.

B-02

Levels of organisation

Answer A

QUESTION 2

Which sequence places the structures in order from **smallest to largest**?

1. circulatory system
2. heart
3. mitochondrion
4. muscle cell
5. muscle tissue

A 3, 4, 5, 2, 1

B 4, 3, 5, 2, 1

C 3, 5, 4, 2, 1

D 3, 4, 2, 5, 1

E 4, 5, 3, 2, 1

WORKED SOLUTION

The hierarchy is:

$$\text{sub-cellular component} < \text{cell} < \text{tissue} < \text{organ} < \text{organ system}$$

Therefore the correct order is **mitochondrion** → **muscle cell** → **muscle tissue** → **heart** → **circulatory system**.

B-03

Gas exchange

Answer **H**

QUESTION 3

Which features increase the rate at which oxygen moves from an alveolus into the blood?

1. A large alveolar surface area.
2. A very thin barrier between alveolar air and blood.
3. Continuous blood flow carrying oxygenated blood away from the alveolus.

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

All three increase diffusion rate. A large surface area provides more area for exchange, a thin barrier shortens diffusion distance, and blood flow removes oxygen that has entered the blood, helping maintain a steep concentration gradient.

B-04

Active transport

Answer C

QUESTION 4

Glucose is being absorbed from the small intestine.

[glucose in intestinal lumen] < [glucose inside an epithelial cell]

but glucose continues to enter the epithelial cell.

Which row best describes this movement and predicts the effect of strongly inhibiting respiration in the epithelial cell?

A diffusion; uptake increases

B osmosis; uptake decreases

C active transport; uptake decreases

D active transport; no effect

E diffusion; no effect

WORKED SOLUTION

Glucose is moving from lower concentration outside the cell to higher concentration inside the cell, so it is moving against its concentration gradient. This requires **active transport**.

Active transport requires energy from respiration, so strongly inhibiting respiration should **decrease uptake**.

B-05 Mitosis and meiosis Answer E

QUESTION 5

A diploid cell contains **8 chromosomes**.

Which row correctly describes the products of mitosis and meiosis?

- **A** mitosis: two cells, 4 chromosomes each; meiosis: four cells, 8 chromosomes each
- **B** mitosis: four cells, 8 chromosomes each; meiosis: two cells, 4 chromosomes each
- **C** mitosis: two cells, 8 chromosomes each; meiosis: two cells, 4 chromosomes each
- **D** mitosis: two cells, 4 chromosomes each; meiosis: four cells, 4 chromosomes each
- **E** mitosis: two cells, 8 chromosomes each; meiosis: four cells, 4 chromosomes each

WORKED SOLUTION

Mitosis produces two daughter cells that retain the diploid chromosome number, so each has **8 chromosomes**.

Meiosis halves the chromosome number and produces four cells, so each has **4 chromosomes**.

B-06

Sex determination

Answer E

QUESTION 6

Assume that the probability of a child being genetically male or female is equal and that the sex of one child does not affect the next.

What is the probability that, in a family's **next three children**, at least one is female?

- **A** 1/8
- **B** 1/4
- **C** 1/2
- **D** 3/4
- **E** 7/8

WORKED SOLUTION

It is easiest to calculate the opposite event: all three children are male.

$$P(\text{all male}) = (1/2)^3 = 1/8$$

$$P(\text{at least one female}) = 1 - 1/8 = \mathbf{7/8}$$

B-07

Inheritance

Answer C

QUESTION 7

In a plant species, tall stems are controlled by a dominant allele **T**, while short stems are controlled by the recessive allele **t**.

Two tall plants produce a short offspring.

What is the probability that their next offspring will also be short?

A 0

B 1/8

C 1/4

D 1/2

E 3/4

WORKED SOLUTION

A short plant must have genotype **tt**. Since two tall plants produced a short offspring, both tall parents must be heterozygous:

$$\mathrm{Tt} \times \mathrm{Tt}$$

The possible genotypes are TT, Tt, Tt and tt, so:

$$P(\mathrm{tt}) = \mathbf{1/4}$$

B-08

DNA base composition

Answer D

QUESTION 8

A double-stranded DNA molecule contains **200 base pairs**.

Cytosine accounts for **22%** of all bases.

How many thymine bases are present?

A 44

B 56

C 88

D 112

E 156

WORKED SOLUTION

There are 400 bases in total. In double-stranded DNA, C and G occur in equal proportions, so:

$$C = 22\%, \quad G = 22\%$$

Thus C + G = 44%, leaving 56% for A + T. Therefore T = 28%.

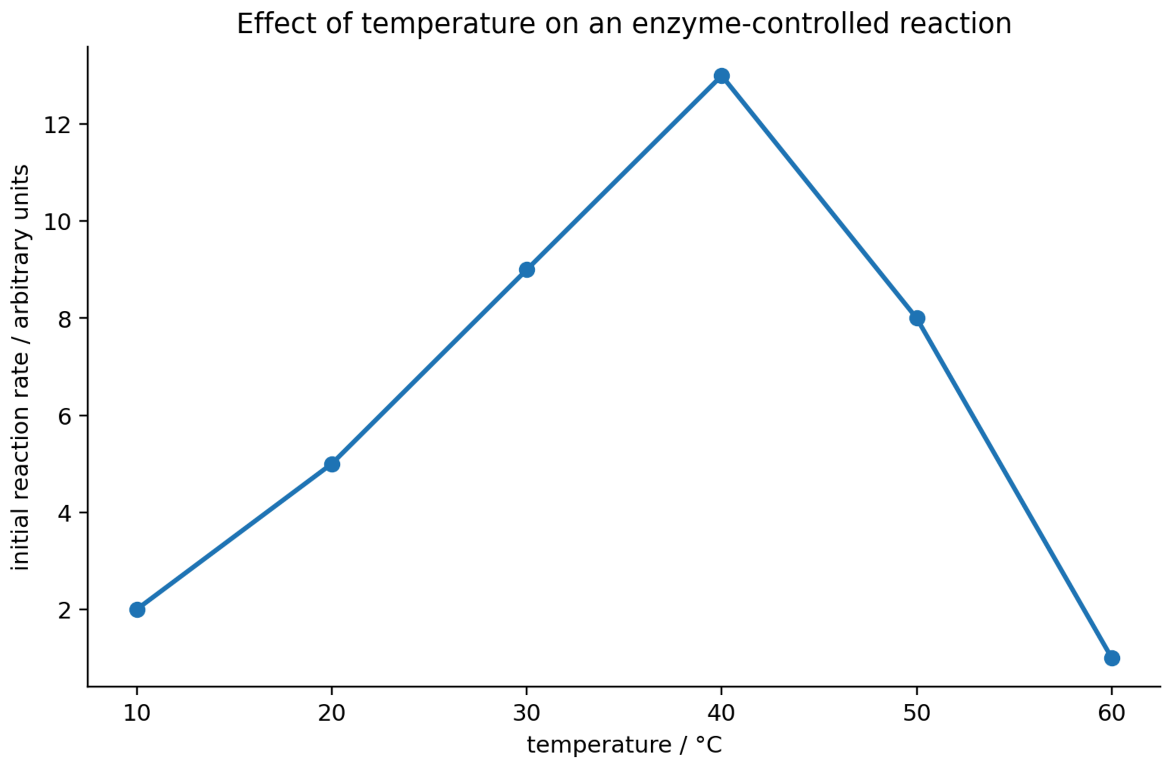
$$0.28 \times 400 = \mathbf{112}$$

B-09

Enzymes Answer **B**

QUESTION 9

The graph shows the initial rate of an enzyme-controlled reaction at six temperatures.



Which statements are justified by these data?

1. The measured reaction rate increases between 10°C and 40°C.
2. The exact optimum temperature of the enzyme must be 40°C.
3. At 60°C, every enzyme molecule must have been permanently denatured.

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Statement 1 is directly supported: the measured rate rises between 10°C and 40°C.

Statement 2 is too strong because temperatures between the measured values were not tested; the true optimum could be slightly below or above 40°C.

Statement 3 is also too strong. The graph still shows a measurable reaction at 60°C, and in any case it does not establish that every enzyme molecule has been permanently denatured.

Therefore **1 only**.

B-10

Digestion | Answer **B**

QUESTION 10

Which row correctly identifies an enzyme, its substrate and one of its products?

- **A** amylase; protein; amino acids
- **B** protease; protein; amino acids
- **C** lipase; starch; simple sugars
- **D** protease; lipid; fatty acids
- **E** lipase; protein; glycerol

WORKED SOLUTION

Proteases catalyse the breakdown of proteins into amino acids. Therefore the correct combination is **protease; protein; amino acids**.

B-11 Contraception Answer E

QUESTION 11

Which statements are correct?

1. Hormonal contraception containing oestrogen and progesterone can inhibit FSH production and help prevent ovulation.
2. Condoms are a non-hormonal form of contraception and can reduce transmission of sexually transmitted infections.
3. Hormonal contraceptives work mainly by directly killing sperm cells.

- **A** none
- **B** 1 only
- **C** 2 only
- **D** 3 only
- **E** 1 and 2 only
- **F** 1 and 3 only
- **G** 2 and 3 only
- **H** 1, 2 and 3

WORKED SOLUTION

Statements 1 and 2 are correct. Hormonal contraception can suppress FSH and therefore help prevent ovulation. Condoms are a barrier method and can also reduce transmission of many sexually transmitted infections.

Hormonal contraception does not principally work by directly killing sperm cells.

B-12

Ciliated cells

Answer G

QUESTION 12

Long-term smoking can damage the cilia lining parts of the respiratory tract.

Which consequences could result directly from reduced ciliary activity?

1. The percentage of oxygen in inhaled air decreases.
2. Mucus may accumulate in the airways.
3. Microorganisms trapped in mucus may be removed less effectively.

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Damaged cilia do not change the percentage of oxygen in the external air being inhaled. However, mucus is cleared less effectively, so it can accumulate and microorganisms trapped in it may remain in the respiratory tract for longer.

Therefore **2 and 3 only**.

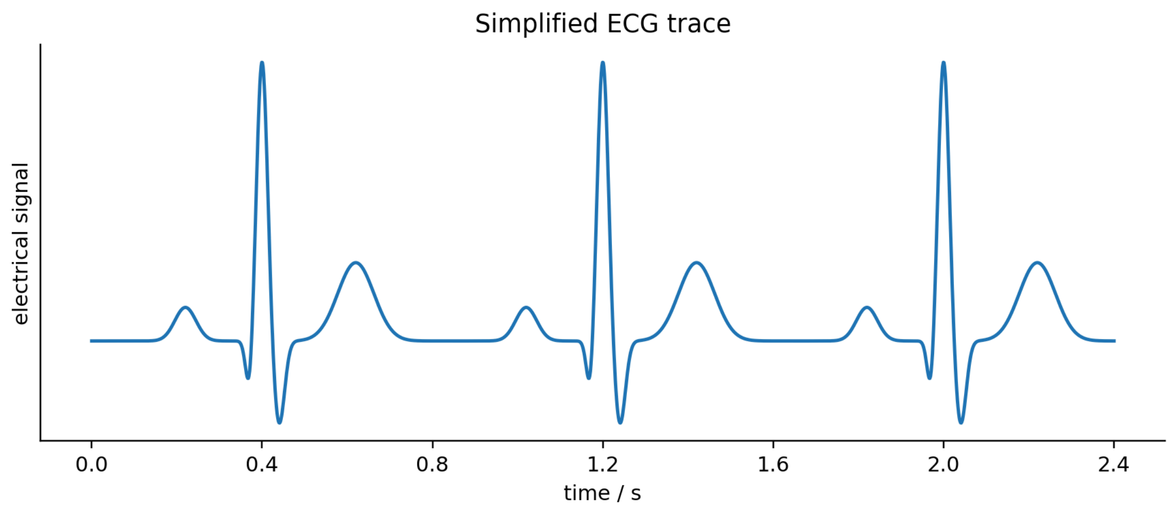
B-13

ECG and heart rate

Answer **C**

QUESTION 13

The trace shows a simplified ECG.



The large R-peaks are approximately **0.80 s** apart.

What is the heart rate?

A 48 beats min^{-1}

B 60 beats min^{-1}

C 75 beats min^{-1}

D 80 beats min^{-1}

E 96 beats min^{-1}

WORKED SOLUTION

One heartbeat takes approximately 0.80 s.

$$\text{heart rate} = 60 / 0.80 = \mathbf{75\ beats\ min^{-1}}$$

B-14

Blood

Answer **D**

QUESTION 14

A patient has:

- a much lower than normal platelet count;
- a much lower than normal red blood cell count.

Which combination of effects is most likely?

A blood clotting faster; oxygen-carrying capacity increased

B blood clotting faster; oxygen-carrying capacity decreased

C blood clotting slower; oxygen-carrying capacity unchanged

D blood clotting slower; oxygen-carrying capacity decreased

E blood clotting unchanged; oxygen-carrying capacity decreased

WORKED SOLUTION

Platelets are required for normal blood clotting, so a low platelet count tends to make clotting slower or less effective.

Red blood cells contain haemoglobin and carry oxygen, so a reduced red blood cell count lowers the blood's oxygen-carrying capacity.

B-15

Respiration | Answer B

QUESTION 15

Four flasks are prepared.

flask	contents	temperature
A	living yeast + glucose solution	20°C
B	living yeast + glucose solution	35°C
C	boiled yeast + glucose solution	35°C
D	living yeast + water only	35°C

Assume 35°C is closer to the optimum temperature for the enzymes involved in yeast respiration than 20°C, and neither temperature causes denaturation.

Which flask is expected to produce the greatest volume of carbon dioxide during the first ten minutes?

A A

B B

C C

D D

E A and B must produce exactly the same amount

WORKED SOLUTION

Flask C contains boiled yeast, so its cells and enzymes will no longer function normally. Flask D lacks glucose.

Both A and B contain living yeast and glucose, but B is at the stated suitable higher temperature of 35°C, so respiration should proceed faster. Therefore **B** should produce the most carbon dioxide in the first ten minutes.

B-16 Thermoregulation Answer **B**

QUESTION 16

A healthy person’s core body temperature rises above its normal value.

Which response is expected?

A sweat production decreases; skin arterioles constrict

B sweat production increases; skin arterioles dilate

C sweat production increases; skin arterioles constrict

D sweat production decreases; skin arterioles dilate

E sweat production unchanged; skin arterioles constrict

WORKED SOLUTION

When body temperature is too high, sweat production increases and skin arterioles dilate. Both changes increase heat loss from the body.

B-17

Synapses | Answer **E**

QUESTION 17

Which statements about a synapse are correct?

1. Neurotransmitters are chemicals.
2. Neurotransmitter molecules can diffuse across the synaptic gap.
3. The electrical impulse itself passes directly across the gap as an electrical current.

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Neurotransmitters are chemical messengers. They are released from one neurone and diffuse across the synaptic gap.

The electrical impulse does not simply travel directly across the gap as an electrical current. Therefore **1 and 2 only**.

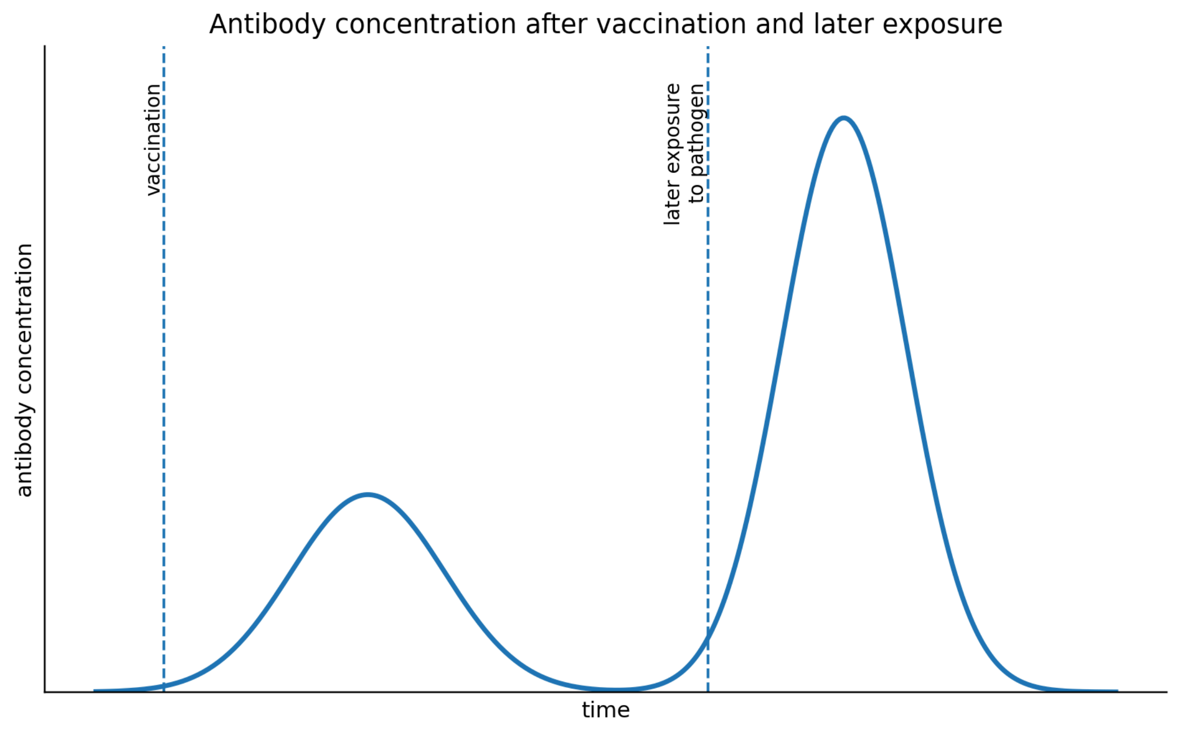
B-18

Vaccination

Answer E

QUESTION 18

The graph shows antibody concentration following vaccination and a later exposure to the same pathogen.



Which statements could explain the graph?

1. Vaccination can lead to the formation of memory cells.
2. Memory cells can allow a faster and larger antibody response during later exposure.
3. The vaccine must contain fully active pathogens capable of producing severe disease.

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Vaccination can generate memory cells. On later exposure to the same antigen, these cells allow a faster and larger antibody response.

A vaccine does not have to contain fully active disease-causing pathogens. Therefore **1 and 2 only**.

B-19 Antibiotic resistance Answer **E**

QUESTION 19

A bacterial population contains a small number of bacteria that are resistant to antibiotic X. Antibiotic X is then used.

Which statements correctly describe natural selection in this population?

1. Resistant bacteria may have been present before the antibiotic was applied.
2. Susceptible bacteria are more likely to die, increasing the relative reproductive success of resistant bacteria.
3. The antibiotic deliberately causes every surviving bacterium to mutate to become resistant.

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

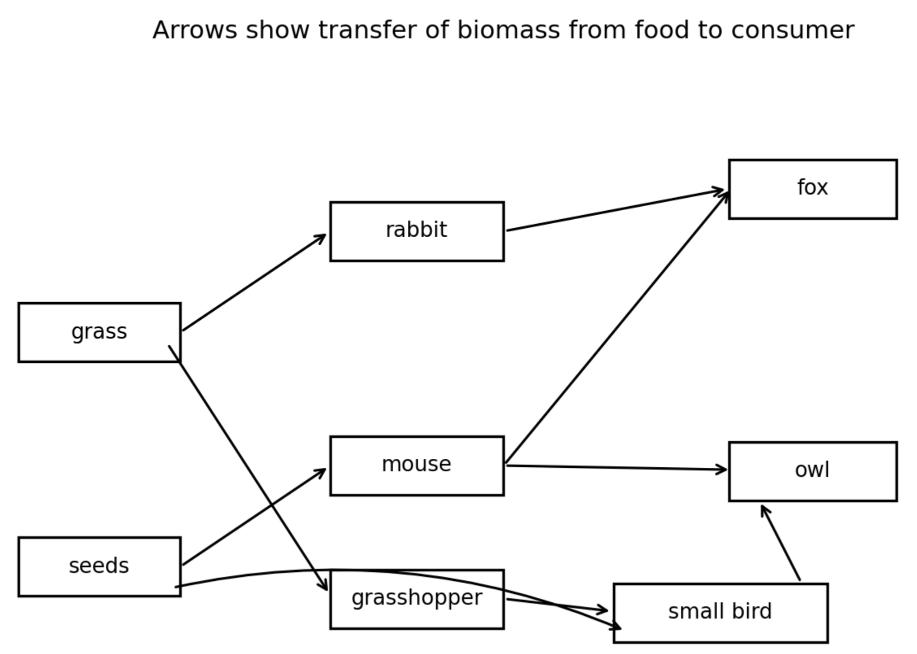
Resistance-conferring variation can exist before treatment. The antibiotic then acts as a selective pressure: susceptible bacteria die more readily while resistant bacteria survive and reproduce.

The antibiotic does not deliberately cause every survivor to mutate. Therefore **1 and 2 only**.

B-20 Food webs Answer E

QUESTION 20

Arrows in the food web show transfer of biomass from food to consumer.



Which organism can act as a **primary consumer and a secondary consumer**, and is also eaten by an owl?

- **A** rabbit
- **B** mouse
- **C** fox
- **D** grasshopper
- **E** small bird

WORKED SOLUTION

The small bird eats seeds, so when doing so it is a primary consumer. It also eats grasshoppers, so it can act as a secondary consumer.

The food web also shows the small bird being eaten by the owl. Therefore the answer is **small bird**.

B-21

Sampling populations

Answer D

QUESTION 21

Five 1.0 m^2 quadrats are placed randomly in a 200 m^2 field.

The number of individuals of plant species X recorded is:

quadrat	number of plants
1	4
2	6
3	5
4	3
5	7

Using these samples, what is the best estimate of the number of plants of species X in the whole field?

A 200

B 400

C 500

D 1000

E 5000

WORKED SOLUTION

The mean number per 1.0 m^2 quadrat is:

$$(4 + 6 + 5 + 3 + 7) / 5 = 25 / 5 = 5$$

Estimated density = 5 plants m^{-2}.

$5 \times 200 =$ **1000 plants**

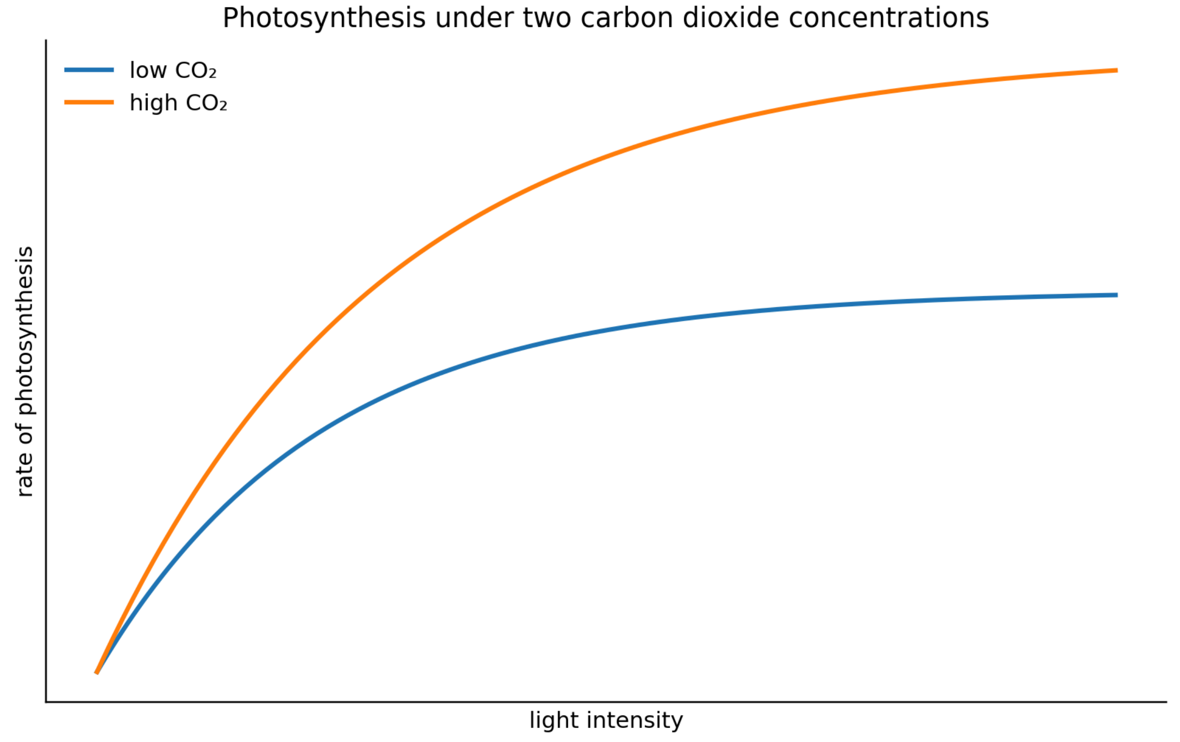
B-22

Limiting factors in photosynthesis

Answer A

QUESTION 22

The graph shows the effect of light intensity under two carbon dioxide concentrations.



Which conclusion is best supported?

A At high light intensity, carbon dioxide concentration can limit the rate of photosynthesis.

B At low carbon dioxide concentration, light intensity can never be limiting.

C Increasing light intensity indefinitely must always increase photosynthesis.

D Carbon dioxide concentration has no effect on photosynthesis.

E Photosynthesis is independent of light intensity.

WORKED SOLUTION

At high light intensity, the low-CO_2 curve levels off below the high-CO_2 curve. Increasing carbon dioxide therefore increases photosynthesis under those conditions.

This supports the conclusion that carbon dioxide concentration can be a limiting factor at high light intensity.

B-23 Plant transport Answer B

QUESTION 23

Which row correctly describes xylem and phloem?

A xylem transports dissolved sugars from leaves; phloem transports water to leaves

B xylem transports water and mineral ions mainly upwards; phloem transports dissolved sugars from leaves to other parts of the plant

C xylem transports sugars only downwards; phloem transports water only upwards

D xylem consists entirely of living cells; phloem consists entirely of dead cells

E xylem transports water by active transport along the vessel; phloem transports only mineral ions

WORKED SOLUTION

Xylem transports water and mineral ions mainly from the roots upwards through the plant.

Phloem transports dissolved sugars produced in leaves to other parts of the plant where they are used or stored.

B-24

Transpiration

Answer C

QUESTION 24

Identical leafy shoots are placed under different environmental conditions.

experiment	temperature	humidity	air movement
A	low	high	still
B	high	low	still
C	high	low	windy
D	low	high	windy

Which experiment would normally be expected to have the **highest rate of transpiration**?

A A

B B

C C

D D

E All four should have equal rates

WORKED SOLUTION

Transpiration generally increases with higher temperature, lower humidity and greater air movement.

Experiment C combines all three: **high temperature, low humidity and wind**. Therefore C should have the highest transpiration rate.

B-25

Human reproduction

Answer **B**

QUESTION 25

Which row correctly identifies the usual site of fertilisation and the usual site at which an embryo implants?

- **A** fertilisation: ovary; implantation: oviduct
- **B** fertilisation: oviduct; implantation: uterus
- **C** fertilisation: uterus; implantation: ovary
- **D** fertilisation: vagina; implantation: uterus
- **E** fertilisation: uterus; implantation: oviduct

WORKED SOLUTION

In humans, fertilisation normally occurs in an **oviduct**. The developing embryo then reaches the uterus and implants in the **uterine lining**.

B-26

Variation

Answer **H**

QUESTION 26

Which statements are correct?

1. Human height is an example of continuous variation.
2. Human ABO blood group is an example of discontinuous variation.

3. Environmental factors such as nutrition can contribute to variation in human height.

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Human height can take many values across a range, so it is continuous variation. ABO blood groups form distinct categories and are therefore discontinuous.

Height is influenced by both genetic and environmental factors, including nutrition. Therefore all three statements are correct.

B-27

Competition

Answer **B**

QUESTION 27

Two plant species are grown either separately or together for the same length of time.

Assume equal starting biomasses of each species are used in the relevant treatments, and all other environmental conditions are kept the same.

treatment	mean final biomass of species A / g	mean final biomass of species B / g
A grown alone	24	—
B grown alone	—	20
A and B grown together	14	12

Which conclusion is best supported by these data?

A Species A and B do not compete.

B Growing together reduces the growth of both species, which is consistent with competition between them.

C Species A competes only for light.

D Species B will always become extinct when grown with A.

E The experiment proves that water is the resource for which the species compete.

WORKED SOLUTION

When grown alone, species A reaches 24 g and species B reaches 20 g. When grown together, both have lower final biomass: 14 g and 12 g respectively.

This is consistent with interspecific competition. However, the data do not identify the particular resource being competed for and do not show that either species must become extinct.

Compilation record: Level 0 uses Engineering Mock Test 1 for Mathematics/Physics and the corresponding Easy Biology/Chemistry source sets.

This solution book is ready to support the later construction of the matching full test.